

Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data*

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First Version: August 31, 2025

This Version: June 6, 2026

Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

*We are extremely grateful to Lawrence Katz, Jesse Shapiro, and Ludwig Straub for their continued guidance and feedback on this project. We are also deeply thankful to David Autor, Erik Brynjolfsson, Gabriel Chodorow-Reich, Oren Danieli, Eric Gold, Jonathon Hazell, Ori Heffetz, David Lagakos, Amanda Pallais, Santiago Medina, and Austin Zheng for their invaluable discussions and advice throughout various stages of this work. We additionally thank Netanel Ben-Porath, Shotaro Beppu, Martin Bernstein, Olena Bogdan, Bharat Chandar, Ruyi Chen, Fiona Chen, Rupsha Debnath, Cameron Deal, Aristotle Epanomeritakis, Sarah Gao, Jay Garg, Shao-Yu Jheng, Xianruo Kang, Iris Li, Saketh Prazad, Ewan Rawcliffe, Emanuel Schertz, Peyman Shahidi, Ragini Srinivasan, James Stratton, and Katherine Wang as well as participants in the *Macro Lunch* at Harvard, *Digital Economy Lab* seminar series at Stanford University, the *Economics of AI Reading Group* at MIT, the *Midwest Economics Association Annual Meetings 2026*, the *Macro Seminar* at the Boston Fed, the *MAD Conference* at Columbia Business School, the *Departmental Seminar* at the Hebrew University of Jerusalem, the *Applied Economics Seminar* at Tel Aviv University, and the *AI in the Workplace Conference* at the University of Chicago for their valuable feedback and discussion. All errors are our own.

Latest Version: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5425555.

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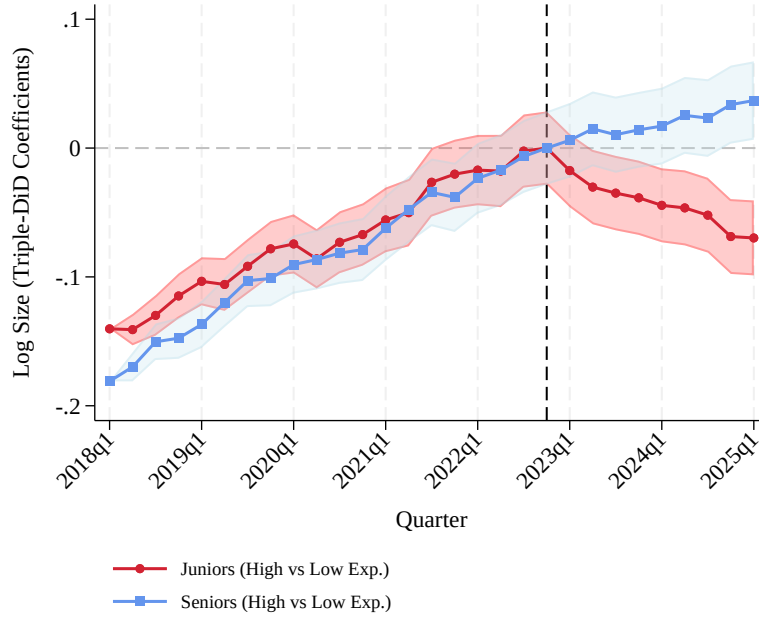


Figure 6: Triple-Difference by Occupational Exposure—Juniors vs. Seniors

Notes: This figure reports the estimated coefficients β_j^s from Equation 12, estimated separately for juniors (red) and seniors (blue). Each series captures the evolution of log employment in high-exposure occupations relative to low-exposure occupations at adopters relative to non-adopters, conditional on firm-by-time and industry-by-exposure-by-time fixed effects. Details on the exposure measure are provided in Section 4.1. Shaded bands are 95% confidence intervals. Standard errors are clustered at the firm level.

estimates the DiD specification within tiers of juniors' educational background, measured by the prestige of their alma mater. The relative decline in junior employment is U-shaped across tiers: largest for juniors from mid-tier institutions and smallest for juniors from the most and least selective institutions.

5.2.4 Discussion: The Timing of the Junior Employment Decline

The early and pronounced decline in the triple-difference coefficients—starting to emerge in 2023Q1, shortly after the release of ChatGPT—may appear surprising, given that automation effects typically unfold more gradually. One plausible explanation is that the rapid diffusion of GenAI tools at the end of 2022 constituted a discrete shock to firms' expectations about future automation. In response, firms may have adjusted in a forward-looking manner by scaling back hiring for junior roles they anticipated would be automated in the near future. Such preemptive adjustments could be optimal for firms if

A.3 Heterogeneity by Educational Background

This appendix examines whether the decline in junior employment varies by workers' educational background, as referenced in Section 5.2.3. We first describe the construction of the school-quality variable, then present the heterogeneity results.

A.3.1 Construction of the School-Quality Variable

To capture educational background, we construct a position-level measure of school quality. For each position, we assign the institution where the worker most recently studied, provided that the education ended no later than one year after the job start. If no such record exists, we use the first recorded institution, provided it began before the job start date. Each institution is then assigned a GPT-4-based quality rating on a five-point scale, produced using OpenAI's GPT-4o-mini model (see Supplemental Appendix A.11.2 for the full prompt). A score of 1 corresponds to Ivy League and other globally elite institutions; 2 to highly respected international universities; 3 to strong national or regional schools; 4 to lower-tier but standard institutions; and 5 to minimally selective or non-accredited institutions.

A.3.2 Results

We re-estimate Equation 34 for junior employment separately within each of the five school-quality tiers. For clarity of presentation, we use a single post-treatment indicator for quarters after 2023Q1 instead of a full set of time dummies.

The results, shown in Figure A.12, reveal a U-shaped pattern. Juniors from tier-3 universities experienced the steepest relative declines in employment, while those from tiers 2 and 4 also saw significant reductions. Juniors from tier-1 and tier-5 universities experienced the smallest decline.³³ Figure A.13 reports the time-series version of these estimates, with a full set of time dummies in place of the single post-2023 indicator.

To help interpret this pattern, Figures A.14 and A.15 show average predicted salaries³⁴

³³In the analysis, we exclude positions where the worker's most recent educational entry is reported as a high school (approximately 5 percent of positions). Including these positions, based on high school quality, has virtually no effect on the results.

³⁴Actual position-level salaries are unavailable. However, Revelio Labs provides a predicted salary